

## ENGINEERING MATHEMATICS III · UNIT III

**Complex Differentiation**

Analytic Functions, Cauchy–Riemann Equations and Harmonic Functions

Class notes of Dr. S. Balaji · August–September 2026

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**1 Complex Functions and Differentiability**

Class of 24/08/26

We know that  $z = x + iy$  is a complex variable and we can construct a complex function  $w = f(z)$  using the complex variable.

**Example**

$z = x + iy$  is a complex variable.

$$f(z) = z^2 = (x + iy)^2 = (x^2 - y^2) + i 2xy$$

$$w = f(z) = u(x, y) + i v(x, y)$$

Like the real function  $y = f(x)$ , we can define the *limit* and *continuity* of  $w = f(z)$ .

**Differentiability**

The derivative of the function  $f(z)$  is defined by

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z},$$

provided the limit exists and is the **same along all the paths** along which  $\Delta z \rightarrow 0$ .

**Example 1:**  $f(z) = z^2$  is differentiable

$$\begin{aligned}
 f(z + \Delta z) &= (z + \Delta z)^2 \\
 \Rightarrow f(z + \Delta z) - f(z) &= (z + \Delta z)^2 - z^2 \\
 &= \cancel{z^2} + (\Delta z)^2 + 2z \Delta z - \cancel{z^2} \\
 &= (\Delta z)^2 + 2z \Delta z
 \end{aligned}$$

$$f(z + \Delta z) - f(z) = \Delta z (\Delta z + 2z)$$

Now

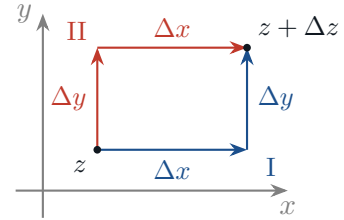
$$\begin{aligned}
 f'(z) &= \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{\cancel{\Delta z} (\Delta z + 2z)}{\cancel{\Delta z}} = \lim_{\Delta z \rightarrow 0} (\Delta z + 2z) \\
 f'(z) &= \lim_{(\Delta x + i\Delta y) \rightarrow 0} [(\Delta x + i\Delta y) + 2z] \quad (1)
 \end{aligned}$$

**Along path I:**  $\Delta y \rightarrow 0$  first and then  $\Delta x \rightarrow 0$ .

$$(1) \Rightarrow f'(z) = \lim_{\Delta x \rightarrow 0} (\Delta x + 2z) \Rightarrow \boxed{f'(z) = 2z} \quad (2)$$

**Along path II:**  $\Delta x \rightarrow 0$  first and then  $\Delta y \rightarrow 0$ .

$$(1) \Rightarrow f'(z) = \lim_{\Delta y \rightarrow 0} (i\Delta y + 2z) \Rightarrow \boxed{f'(z) = 2z} \quad (3)$$



By (2) and (3),  $f'(z)$  is the same along the two paths where  $\Delta z \rightarrow 0$ .

$$\therefore f(z) = z^2 \text{ is a differentiable function and } f'(z) = 2z.$$

**Example 2:**  $f(z) = \bar{z}$  is not differentiable

$$\begin{aligned}
 f(z + \Delta z) &= \overline{z + \Delta z} \\
 f(z + \Delta z) - f(z) &= \overline{(z + \Delta z)} - \bar{z} \\
 &= \overline{(x + iy + \Delta x + i\Delta y)} - \overline{(x + iy)} \\
 &= \overline{((x + \Delta x) + i(y + \Delta y))} - (x - iy) \\
 &= (\cancel{x} + \Delta x - i(y + \Delta y)) - (\cancel{x} - \cancel{iy}) \\
 f(z + \Delta z) - f(z) &= \Delta x - i\Delta y
 \end{aligned}$$

Then

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{(\Delta x + i\Delta y) \rightarrow 0} \frac{\Delta x - i\Delta y}{\Delta x + i\Delta y} \quad (1)$$

**Along path I:** first  $\Delta y \rightarrow 0$ , then  $\Delta x \rightarrow 0$ .

$$(1) \Rightarrow f'(z) = \lim_{\Delta x \rightarrow 0} \frac{\Delta x}{\Delta x} \Rightarrow \boxed{f'(z) = 1} \quad (2)$$

**Along path II:** first  $\Delta x \rightarrow 0$ , then  $\Delta y \rightarrow 0$ .

$$(1) \Rightarrow f'(z) = \lim_{\Delta y \rightarrow 0} \frac{-i\Delta y}{i\Delta y} \Rightarrow \boxed{f'(z) = -1} \quad (3)$$

We see that the limits along the two paths are not the same.

∴ By (2) and (3),  $f(z) = \bar{z}$  is **not** a differentiable function.

## 2 Analytic Functions and the Cauchy–Riemann Equations

### Analytic function

A complex function  $f(z)$  is said to be **analytic** at  $z = z_0$  if  $f(z)$  is differentiable at  $z = z_0$  and also differentiable at every point of some neighbourhood of  $z_0$ .

If  $f(z)$  is differentiable at each and every point of a region  $R$ , then we say that  $f(z)$  is **analytic in the region  $R$** .

### Cauchy–Riemann (C–R) equations

The necessary condition for a function  $f(z)$  to be analytic is given by: if  $f(z) = u(x, y) + iv(x, y)$  is analytic, then it satisfies the equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

i.e.  $u_x = v_y$  and  $u_y = -v_x$

### Example: $f(z) = z^2$

$f(z) = z^2$  can be written as  $f(z) = (x + iy)^2 = (x^2 - y^2) + i2xy = u + iv$ .

$$\begin{aligned} u &= x^2 - y^2 \\ \frac{\partial u}{\partial x} &= 2x \\ \frac{\partial u}{\partial y} &= -2y \end{aligned}$$

$$\begin{aligned} v &= 2xy \\ \frac{\partial v}{\partial x} &= 2y \\ \frac{\partial v}{\partial y} &= 2x \end{aligned}$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

The C–R equations are satisfied;  $f(z) = z^2$  is an analytic function.

### 2.1 Proof of the C–R equations

Class of 31/08/26

Let  $f(z) = u + iv = u(x, y) + i v(x, y)$  be an analytic function. Then

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \quad \text{--- (1)}$$

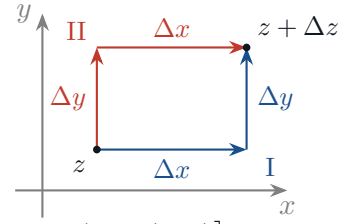
exists and is independent of the path along which  $\Delta z \rightarrow 0$ .

Now  $f(z) = u(x, y) + iv(x, y)$  and  $f(z + \Delta z) = u(x + \Delta x, y + \Delta y) + i v(x + \Delta x, y + \Delta y)$ .

$$\begin{aligned} f(z + \Delta z) - f(z) &= [u(x + \Delta x, y + \Delta y) + i v(x + \Delta x, y + \Delta y)] - [u(x, y) + i v(x, y)] \\ &= [u(x + \Delta x, y + \Delta y) - u(x, y)] + i[v(x + \Delta x, y + \Delta y) - v(x, y)] \end{aligned}$$

$$\therefore f'(z) = \lim_{(\Delta x + i\Delta y) \rightarrow 0} \frac{[u(x + \Delta x, y + \Delta y) - u(x, y)] + i[v(x + \Delta x, y + \Delta y) - v(x, y)]}{\Delta x + i\Delta y} \quad \text{--- (2)}$$

**Case (i): Along path I** — first  $\Delta y \rightarrow 0$  and then  $\Delta x \rightarrow 0$ .



$$\begin{aligned}
 (2) \implies f'(z) &= \lim_{\Delta x \rightarrow 0} \frac{[u(x + \Delta x, y) - u(x, y)] + i[v(x + \Delta x, y) - v(x, y)]}{\Delta x} \\
 &= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \lim_{\Delta x \rightarrow 0} \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x} \\
 f'(z) &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \quad \text{--- (3)}
 \end{aligned}$$

**Case (ii): Along path II** — first  $\Delta x \rightarrow 0$  and then  $\Delta y \rightarrow 0$ .

$$\begin{aligned}
 (2) \implies f'(z) &= \lim_{\Delta y \rightarrow 0} \frac{[u(x, y + \Delta y) - u(x, y)] + i[v(x, y + \Delta y) - v(x, y)]}{i \Delta y} \\
 &= \lim_{\Delta y \rightarrow 0} \frac{u(x, y + \Delta y) - u(x, y)}{i \Delta y} + \lim_{\Delta y \rightarrow 0} \frac{v(x, y + \Delta y) - v(x, y)}{\Delta y} \\
 f'(z) &= -i \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} \quad \text{--- (4)} \quad \left( \because \frac{1}{i} = -i \right)
 \end{aligned}$$

Since  $f(z)$  is analytic,  $f'(z)$  along any path must be the same.  $\therefore$  (3) = (4):

$$\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}$$

Equating real and imaginary parts,

#### C-R equations (Cartesian form)

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad \text{i.e.} \quad u_x = v_y, \quad u_y = -v_x$$

Hence the C-R equations.

## 2.2 C-R equations in polar form

If we put  $z = re^{i\theta}$ , we get the complex function  $f(z)$  in polar form,  $f(z) = u(r, \theta) + i v(r, \theta)$ . In this case the C-R equations in polar form are:

#### C-R equations (polar form)

$$\begin{aligned}
 \frac{\partial u}{\partial r} &= \frac{1}{r} \frac{\partial v}{\partial \theta} \quad \text{and} \quad \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta} \\
 \text{i.e.} \quad u_r &= \frac{1}{r} v_\theta \quad \text{and} \quad v_r = -\frac{1}{r} u_\theta
 \end{aligned}$$

## 3 Problems on Analyticity

**Problem 1**

Check the analyticity of the following functions:

(i)  $f(z) = \sin z$

(iii)  $f(z) = \cosh z$

(v)  $f(z) = z^3$

(ii)  $f(z) = xy + iy$

(iv)  $f(z) = e^z$

(vi)  $f(z) = z^n$

**Useful results**

$$\sin(ix) = i \sinh x, \quad \cos(ix) = \cosh x, \quad e^{iy} = \cos y + i \sin y$$

**Solution**

(i) Given  $f(z) = \sin z$ .

$$\begin{aligned} f(z) &= \sin(x + iy) = \sin x \cos(iy) + \cos x \sin(iy) \\ &= \sin x \cosh y + i \cos x \sinh y \\ &= u(x, y) + i v(x, y) \end{aligned}$$

$$\begin{aligned} u &= \sin x \cosh y \\ \frac{\partial u}{\partial x} &= \cos x \cosh y \\ \frac{\partial u}{\partial y} &= \sin x \sinh y \end{aligned}$$

$$\begin{aligned} v &= \cos x \sinh y \\ \frac{\partial v}{\partial x} &= -\sin x \sinh y \\ \frac{\partial v}{\partial y} &= \cos x \cosh y \end{aligned}$$

$$\Rightarrow \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}, \quad \text{i.e. } u_x = v_y \text{ and } u_y = -v_x.$$

C-R equations are satisfied.  $\therefore f(z) = \sin z$  is an analytic function.

(ii) Given  $f(z) = xy + iy = u + iv$ .

$$\begin{aligned} u &= xy \\ u_x &= y \\ u_y &= x \end{aligned}$$

$$\begin{aligned} v &= y \\ v_x &= 0 \\ v_y &= 1 \end{aligned}$$

$u_x \neq v_y$  and  $u_y \neq -v_x$   
C-R equations are not satisfied.

$\therefore f(z) = xy + iy$  is **not** an analytic function.

(iii) Given  $f(z) = \cosh z$ . Let

$$\begin{aligned} \cosh z &= \cos(iz) = \cos(i(x + iy)) = \cos(ix - y) \\ &= \cos(ix) \cos y + \sin(ix) \sin y \\ &= \cosh x \cos y + i \sinh x \sin y \\ &= u(x, y) + i v(x, y) \end{aligned}$$

$$\begin{aligned} u &= \cosh x \cos y \\ \frac{\partial u}{\partial x} &= \sinh x \cos y \\ \frac{\partial u}{\partial y} &= -\cosh x \sin y \end{aligned}$$

$$\begin{aligned} v &= \sinh x \sin y \\ \frac{\partial v}{\partial x} &= \cosh x \sin y \\ \frac{\partial v}{\partial y} &= \sinh x \cos y \end{aligned}$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}, \quad \text{i.e. } u_x = v_y \text{ and } v_x = -u_y.$$

C-R equations are satisfied.  $\therefore f(z) = \cosh z$  is an analytic function.

(iv) Given  $f(z) = e^z$ .

$$f(z) = e^{x+iy} \implies f(z) = e^x e^{iy} = e^x (\cos y + i \sin y) \quad (\because e^{iy} = \cos y + i \sin y)$$

$$\begin{aligned} u &= e^x \cos y \\ \frac{\partial u}{\partial x} &= e^x \cos y \\ \frac{\partial u}{\partial y} &= -e^x \sin y \end{aligned}$$

$$\begin{aligned} v &= e^x \sin y \\ \frac{\partial v}{\partial x} &= e^x \sin y \\ \frac{\partial v}{\partial y} &= e^x \cos y \end{aligned}$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

C-R equations are satisfied.  $\therefore f(z) = e^z$  is an analytic function.

(v) Given  $f(z) = z^3$ .

$$\begin{aligned} f(z) &= (x + iy)^3 = x^3 + 3x^2(iy) + 3x(iy)^2 + (iy)^3 \\ &= x^3 + i3x^2y - 3xy^2 - iy^3 \\ &= (x^3 - 3xy^2) + i(3x^2y - y^3) \end{aligned}$$

$$\begin{aligned} u &= x^3 - 3xy^2 \\ \frac{\partial u}{\partial x} &= 3x^2 - 3y^2 \\ \frac{\partial u}{\partial y} &= -6xy \end{aligned}$$

$$\begin{aligned} v &= 3x^2y - y^3 \\ \frac{\partial v}{\partial x} &= 6xy \\ \frac{\partial v}{\partial y} &= 3x^2 - 3y^2 \end{aligned}$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 3x^2 - 3y^2, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = -6xy$$

C-R equations are satisfied.  $\therefore f(z) = z^3$  is an analytic function.

(vi) Given  $f(z) = z^n$ . Using the polar form  $z = re^{i\theta}$ ,

$$\begin{aligned} f(z) &= (re^{i\theta})^n = r^n e^{in\theta} = r^n (\cos n\theta + i \sin n\theta) \\ &= r^n \cos n\theta + i r^n \sin n\theta = u(r, \theta) + i v(r, \theta) \end{aligned}$$

$$\begin{array}{lcl}
 u = r^n \cos n\theta & & v = r^n \sin n\theta \\
 \frac{\partial u}{\partial r} = n r^{n-1} \cos n\theta & & \frac{\partial v}{\partial r} = n r^{n-1} \sin n\theta \\
 \frac{\partial u}{\partial \theta} = -n r^n \sin n\theta & & \frac{\partial v}{\partial \theta} = n r^n \cos n\theta \\
 -\frac{1}{r} \frac{\partial u}{\partial \theta} = n r^{n-1} \sin n\theta = \frac{\partial v}{\partial r} & & \frac{1}{r} \frac{\partial v}{\partial \theta} = n r^{n-1} \cos n\theta = \frac{\partial u}{\partial r} \\
 \therefore \frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} & \text{and} & \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}
 \end{array}$$

C-R equations in polar form are satisfied.  $\therefore f(z) = z^n$  is an analytic function.

### Problem 2

Prove that  $f(z) = |z|^2$  is differentiable only at the origin  $(0,0)$  and is not analytic anywhere. *Class of 02/09/26*

**Solution.** Given  $f(z) = |z|^2 = x^2 + y^2 = u + iv$ , so  $u = x^2 + y^2$  and  $v = 0$ .

$$\begin{array}{lcl}
 \frac{\partial u}{\partial x} = 2x & & \frac{\partial v}{\partial x} = 0 \\
 \frac{\partial u}{\partial y} = 2y & & \frac{\partial v}{\partial y} = 0
 \end{array}$$

$\Rightarrow$  C-R equations are satisfied only when  $x = 0$  and  $y = 0$ . But in the neighbourhood of  $(0,0)$ , the C-R equations are not satisfied.

$\Rightarrow f(z) = |z|^2$  is differentiable only at the origin and is not analytic anywhere.

## 4 Properties of Analytic Functions

### Property (i)

An analytic function with constant real part is constant.

### Proof

Let  $f(z) = u + iv$  be an analytic function with  $u = c$ , a constant. — (1)

Since  $f(z) = u + iv$  is analytic, it satisfies the C-R equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\Rightarrow 0 = \frac{\partial v}{\partial y} \quad \text{and} \quad 0 = \frac{\partial v}{\partial x} \quad (\because u = c) \quad \text{i.e. } v_x = 0 \text{ and } v_y = 0,$$

which is possible only when  $v = k$ , a constant.

$$\Rightarrow f(z) = u + iv = c + ik$$

$f(z)$  is a complex constant.

**Note**

Similarly we can prove that an analytic function with constant imaginary part is also constant.

**Property (ii)**

An analytic function with constant modulus is also a constant function.

**Proof**

Given  $f(z) = u + iv$  is an analytic function with constant modulus.

$$\begin{aligned} \implies |f(z)| &= \sqrt{u^2 + v^2} = \text{a constant} \\ \implies u^2 + v^2 &= \text{a constant} = c \text{ (say)} \\ \implies u^2 + v^2 &= c \end{aligned} \quad \text{--- (1)}$$

Differentiating (1) partially w.r.t.  $x$ :

$$2uu_x + 2vv_x = 0 \implies uu_x + vv_x = 0 \quad \text{--- (2)}$$

Differentiating (1) partially w.r.t.  $y$ :

$$2uu_y + 2vv_y = 0 \implies uu_y + vv_y = 0$$

By the C-R equations ( $u_y = -v_x$ ,  $v_y = u_x$ ):

$$u(-v_x) + v(u_x) = 0 \implies vu_x - uv_x = 0 \quad \text{--- (3)}$$

By (2) and (3),

$$\begin{cases} uu_x + vv_x = 0 \\ vu_x - uv_x = 0 \end{cases} \quad \begin{vmatrix} u & v \\ v & -u \end{vmatrix} = -u^2 - v^2 = -(u^2 + v^2) = -c \neq 0$$

This is a system of homogeneous equations in  $u_x$  and  $v_x$  whose determinant value is non-zero.

$\implies$  The system possesses only the zero solution

$\implies u_x = 0$  and  $v_x = 0$

$\implies v_y = 0$  and  $u_y = 0$  ( $\because$  by C-R equations)

$$u_x = u_y = 0 \implies u = c_1, \text{ a constant}; \quad v_x = v_y = 0 \implies v = c_2, \text{ a constant}$$

$$f(z) = u + iv = c_1 + ic_2$$

$f(z)$  is a complex constant.

**Property (iii)**

In an analytic function  $f(z) = u + iv$ , the family of curves  $u(x, y) = c_1$  and  $v(x, y) = c_2$  intersect orthogonally.

**Proof**

Given  $f(z) = u + iv$  is an analytic function  $\implies$  the C-R equations are satisfied:

$$u_x = v_y \quad \text{and} \quad u_y = -v_x \quad \text{--- (1)}$$

Consider the family of curves  $u(x, y) = c_1$ , a constant. Then by the total derivative

$$\begin{aligned} du &= \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \\ \Rightarrow 0 &= \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \Rightarrow \frac{\partial u}{\partial y} dy = -\frac{\partial u}{\partial x} dx \\ \frac{dy}{dx} &= \frac{-\partial u / \partial x}{\partial u / \partial y} \Rightarrow m_1 = -\frac{u_x}{u_y} \quad \text{--- (2)} \end{aligned}$$

Now consider the family  $v(x, y) = c_2$ :

$$\begin{aligned} dv &= \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy \\ \Rightarrow 0 &= \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy \Rightarrow \frac{\partial v}{\partial y} dy = -\frac{\partial v}{\partial x} dx \\ \frac{dy}{dx} &= -\frac{v_x}{v_y} \Rightarrow m_2 = -\frac{v_x}{v_y} \quad \text{--- (3)} \end{aligned}$$

Then

$$m_1 \cdot m_2 = \left(-\frac{u_x}{u_y}\right) \times \left(-\frac{v_x}{v_y}\right) = \frac{u_x}{u_y} \times \left(\frac{-u_y}{u_x}\right) = -1 \quad (\because \text{by C-R equations})$$

$\Rightarrow$  The families of curves  $u = c_1$  and  $v = c_2$  cut orthogonally.

### Property (iv)

The real and imaginary parts of an analytic function satisfy Laplace's equation, i.e.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{and} \quad \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0.$$

### Proof

By the C-R equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{--- (1)} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad \text{--- (2)}$$

Differentiate (1) w.r.t.  $x$ :

$$\begin{aligned} \frac{\partial}{\partial x} \left( \frac{\partial u}{\partial x} \right) &= \frac{\partial}{\partial x} \left( \frac{\partial v}{\partial y} \right) \\ \frac{\partial^2 u}{\partial x^2} &= \frac{\partial^2 v}{\partial x \partial y} \end{aligned}$$

Differentiate (2) w.r.t.  $y$ :

$$\begin{aligned} \frac{\partial}{\partial y} \left( \frac{\partial u}{\partial y} \right) &= -\frac{\partial}{\partial y} \left( \frac{\partial v}{\partial x} \right) \\ \frac{\partial^2 u}{\partial y^2} &= -\frac{\partial^2 v}{\partial y \partial x} \end{aligned}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 v}{\partial x \partial y} - \frac{\partial^2 v}{\partial y \partial x} \quad \therefore \quad \boxed{\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0}$$

Similarly, differentiate (1) w.r.t.  $y$ :

$$\frac{\partial}{\partial y} \left( \frac{\partial u}{\partial x} \right) = \frac{\partial}{\partial y} \left( \frac{\partial v}{\partial y} \right)$$

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 v}{\partial y^2}$$

Similarly, differentiate (2) w.r.t.  $x$ :

$$-\frac{\partial}{\partial x} \left( \frac{\partial v}{\partial x} \right) = \frac{\partial}{\partial x} \left( \frac{\partial u}{\partial y} \right)$$

$$-\frac{\partial^2 v}{\partial x^2} = \frac{\partial^2 u}{\partial x \partial y}$$

$$\frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial x^2} = \frac{\partial^2 u}{\partial y \partial x} - \frac{\partial^2 u}{\partial x \partial y} \quad \therefore \quad \boxed{\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0}$$

### Problem 3

Prove that an analytic function with constant argument is constant.

### Proof

Let  $f(z) = u(x, y) + iv(x, y) = \rho(x, y) e^{ic}$ , where the argument  $c$  is a constant.

$$u = \rho \cos c$$

$$\frac{\partial u}{\partial x} = \frac{\partial \rho}{\partial x} \cos c$$

$$\frac{\partial u}{\partial y} = \frac{\partial \rho}{\partial y} \cos c$$

$$v = \rho \sin c$$

$$\frac{\partial v}{\partial x} = \frac{\partial \rho}{\partial x} \sin c$$

$$\frac{\partial v}{\partial y} = \frac{\partial \rho}{\partial y} \sin c$$

As  $f(z)$  is analytic,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \implies \frac{\partial \rho}{\partial x} \cos c = \frac{\partial \rho}{\partial y} \sin c \implies \rho_x \cos c - \rho_y \sin c = 0 \quad \text{--- (1)}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \implies \frac{\partial \rho}{\partial y} \cos c = -\frac{\partial \rho}{\partial x} \sin c \implies \rho_x \sin c + \rho_y \cos c = 0 \quad \text{--- (2)}$$

$$\begin{pmatrix} \cos c & -\sin c \\ \sin c & \cos c \end{pmatrix} \begin{pmatrix} \rho_x \\ \rho_y \end{pmatrix} = 0, \quad \left| \begin{pmatrix} \cos c & -\sin c \\ \sin c & \cos c \end{pmatrix} \right| = \cos^2 c + \sin^2 c = 1 \neq 0$$

Since the determinant is non-zero,  $\rho_x = \rho_y = 0$ . Hence  $\rho$  is a constant, and so  $f(z) = \rho e^{ic}$  is a constant.

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### Notes

(i) By the above result (Property (iv)) we can say that  $u$  and  $v$  are **harmonic functions**, and they are also called **harmonic conjugates** of each other for an analytic function  $f(z) = u + iv$ .

(ii) For the analytic function  $f(z) = u + iv$ , from the derivation of the C-R equations, we can write

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \quad (\text{or}) \quad f'(z) = -i \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}$$

$$\text{i.e. } f'(z) = u_x + iv_x \quad (\text{or}) \quad f'(z) = -iu_y + v_y$$

$$\implies |f'(z)|^2 = u_x^2 + v_x^2 \quad (\text{or}) \quad |f'(z)|^2 = u_y^2 + v_y^2.$$

## 5 Identities for Analytic Functions

**Identity (i)**

For an analytic function  $f(z) = u + iv$ ,

$$\left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |f(z)|^2 = 4 |f'(z)|^2.$$

**Proof**

We know that  $f(z) = u + iv$  is an analytic function  $\implies$  it satisfies the C-R equations.

$$\begin{aligned} \implies |f(z)|^2 &= u^2 + v^2 \\ \frac{\partial}{\partial x} |f(z)|^2 &= \frac{\partial}{\partial x} (u^2 + v^2) = 2uu_x + 2vv_x = 2(uu_x + vv_x) \\ \frac{\partial^2}{\partial x^2} |f(z)|^2 &= 2 \frac{\partial}{\partial x} (uu_x + vv_x) = 2[u_x u_x + uu_{xx} + v_x v_x + vv_{xx}] \\ \frac{\partial^2}{\partial x^2} |f(z)|^2 &= 2[u_x^2 + v_x^2 + uu_{xx} + vv_{xx}] \end{aligned} \quad \text{--- (1)}$$

Similarly,

$$\frac{\partial^2}{\partial y^2} |f(z)|^2 = 2[u_y^2 + v_y^2 + uu_{yy} + vv_{yy}] \quad \text{--- (2)}$$

(1) + (2)  $\implies$

$$\begin{aligned} \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |f(z)|^2 &= 2[u_x^2 + v_x^2 + u_y^2 + v_y^2 + u(u_{xx} + u_{yy}) + v(v_{xx} + v_{yy})] \\ &= 2[|f'(z)|^2 + |f'(z)|^2 + u(0) + v(0)] \quad (\because \text{by Laplace's equation}) \\ &= 2[2|f'(z)|^2] \end{aligned}$$

$$\left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |f(z)|^2 = 4 |f'(z)|^2$$

**Problem 1**

Prove that  $\left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |\operatorname{Re} f(z)|^2 = 2 |f'(z)|^2$  if  $f(z)$  is an analytic function.

**Proof**

We know that  $f(z) = u + iv$  is an analytic function  $\implies$  it satisfies the C-R equations.

$$\begin{aligned} |\operatorname{Re} f(z)|^2 &= u^2 \\ \frac{\partial}{\partial x} |\operatorname{Re} f(z)|^2 &= \frac{\partial}{\partial x} (u^2) = 2uu_x \\ \frac{\partial^2}{\partial x^2} |\operatorname{Re} f(z)|^2 &= 2 \frac{\partial}{\partial x} (uu_x) = 2[u_x u_x + uu_{xx}] \\ \frac{\partial^2}{\partial x^2} |\operatorname{Re} f(z)|^2 &= 2[u_x^2 + uu_{xx}] \end{aligned} \quad \text{--- (1)}$$

Similarly,

$$\frac{\partial^2}{\partial y^2} |\operatorname{Re} f(z)|^2 = 2[u_y^2 + uu_{yy}] \quad \text{--- (2)}$$

(1) + (2)  $\Rightarrow$

$$\begin{aligned} \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |\operatorname{Re} f(z)|^2 &= 2[(u_x^2 + u_y^2) + u(u_{xx} + u_{yy})] \\ &= 2|f'(z)|^2 + u(0) \quad (\because u_y = -v_x \Rightarrow u_x^2 + u_y^2 = u_x^2 + v_x^2 = |f'(z)|^2) \end{aligned}$$

$$\boxed{\left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |\operatorname{Re} f(z)|^2 = 2|f'(z)|^2}$$

### Problem 2 (★)

If  $f(z)$  is analytic, prove that  $\left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \log_e |f(z)| = 0$ .

### Proof

Given  $f(z) = u + iv$  is an analytic function  $\Rightarrow$  it satisfies the C-R equations.

$$|f(z)| = \sqrt{u^2 + v^2} \quad \Rightarrow \quad \log_e |f(z)| = \frac{1}{2} \log_e (u^2 + v^2)$$

$$\frac{\partial}{\partial x} (\log_e |f(z)|) = \frac{1}{2} \frac{\partial}{\partial x} (\log_e (u^2 + v^2)) = \frac{1}{2} \left( \frac{1}{u^2 + v^2} (2uu_x + 2vv_x) \right)$$

$$\frac{\partial}{\partial x} (\log_e |f(z)|) = \frac{uu_x + vv_x}{u^2 + v^2}$$

$$\begin{aligned} \frac{\partial^2}{\partial x^2} (\log_e |f(z)|) &= \frac{\partial}{\partial x} \left[ \frac{uu_x + vv_x}{u^2 + v^2} \right] \\ &= \frac{(u^2 + v^2)(uu_{xx} + vv_{xx}) - (uu_x + vv_x)(2uu_x + 2vv_x)}{(u^2 + v^2)^2} \\ &= \frac{(u^2 + v^2)(u_x^2 + uu_{xx} + v_x^2 + vv_{xx}) - 2(u^2 u_x^2 + 2uv u_x v_x + v^2 v_x^2)}{(u^2 + v^2)^2} \end{aligned}$$

$$\begin{aligned} \frac{\partial^2}{\partial x^2} (\log_e |f(z)|) &= \frac{1}{(u^2 + v^2)^2} \left[ (u^2 + v^2)(u_x^2 + v_x^2 + uu_{xx} + vv_{xx}) \right. \\ &\quad \left. - 2(u^2 u_x^2 + 2uv u_x v_x + v^2 v_x^2) \right] \quad \text{--- (1)} \end{aligned}$$

Similarly,

$$\begin{aligned} \frac{\partial^2}{\partial y^2} (\log_e |f(z)|) &= \frac{1}{(u^2 + v^2)^2} \left[ (u^2 + v^2)(u_y^2 + v_y^2 + uu_{yy} + vv_{yy}) \right. \\ &\quad \left. - 2(u^2 u_y^2 + 2uv u_y v_y + v^2 v_y^2) \right] \quad \text{--- (2)} \end{aligned}$$

(1) + (2)  $\Rightarrow$

$$\begin{aligned} &\left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \log_e |f(z)| \\ &= \frac{1}{(u^2 + v^2)^2} \left[ (u^2 + v^2) \{ (u_x^2 + v_x^2) + (u_y^2 + v_y^2) + u(u_{xx} + u_{yy}) + v(v_{xx} + v_{yy}) \} \right] \end{aligned}$$

$$\begin{aligned}
& -2\{u^2(u_x^2 + u_y^2) + 2uv(u_x v_x + u_y v_y) + v^2(v_x^2 + v_y^2)\} \\
&= \frac{1}{(u^2 + v^2)^2} \left[ (u^2 + v^2) \{ |f'(z)|^2 + |f'(z)|^2 + 0 + 0 \} \right. \\
&\quad \left. - 2\{u^2 |f'(z)|^2 + 0 + v^2 |f'(z)|^2\} \right] \\
&= \frac{1}{(u^2 + v^2)^2} \left[ 2(u^2 + v^2) |f'(z)|^2 - 2(u^2 + v^2) |f'(z)|^2 \right] = 0
\end{aligned}$$

(Here  $u_x v_x + u_y v_y = u_x v_x + (-v_x)(u_x) = 0$  by the C-R equations.)

$$\therefore \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \log_e |f(z)| = 0$$

## 6 Construction of Analytic Functions — Milne–Thomson Method

Class of 07/09/26

We can construct the analytic function  $f(z) = u + iv$  when either its real part  $u$  or its imaginary part  $v$  is given.

**Milne–Thomson method.** Here we follow the procedure of the Milne–Thomson method together with the  $f'(z)$  obtained from the derivation of the C–R equations.

### Case 1: real part $u$ is given

$$f(z) = \int (\phi_1 - i \phi_2) dz + c, \quad \text{where } \phi_1 = \frac{\partial u}{\partial x}(z, 0), \quad \phi_2 = \frac{\partial u}{\partial y}(z, 0)$$

### Case 2: imaginary part $v$ is given

$$f(z) = \int (\psi_1 + i \psi_2) dz + c, \quad \text{where } \psi_1 = \frac{\partial v}{\partial y}(z, 0), \quad \psi_2 = \frac{\partial v}{\partial x}(z, 0)$$

### Bernoulli's formula (integration by parts)

$$\int u v dz = u v_1 - u' v_2 + u'' v_3 - \dots$$

where  $v_1, v_2, \dots$  are the successive integrals of  $v$  and  $u', u'', \dots$  the successive derivatives of  $u$ .

### Problem 1 (★)

Construct the analytic function  $f(z) = u + iv$  by the Milne–Thomson method, given  $v = e^{-x}(x \cos y + y \sin y)$ .

**Solution.** Given  $v = e^{-x}(x \cos y + y \sin y)$ .

$$\begin{aligned}
\frac{\partial v}{\partial x} &= e^{-x}(\cos y + 0) + (-e^{-x})(x \cos y + y \sin y) \\
\frac{\partial v}{\partial x} &= e^{-x} \cos y - e^{-x}(x \cos y + y \sin y) \\
\psi_2 &= \frac{\partial v}{\partial x}(z, 0) = e^{-z} \cos 0 - e^{-z}(z(1) + 0) = e^{-z} - z e^{-z}
\end{aligned}$$

$$\psi_2 = (1 - z) e^{-z}$$

Now

$$\begin{aligned}\frac{\partial v}{\partial y} &= e^{-x}(x(-\sin y) + y \cos y + 1 \cdot \sin y) = e^{-x}((1 - x) \sin y + y \cos y) \\ \psi_1 &= \frac{\partial v}{\partial y}(z, 0) = e^{-z}((1 - z)(0) + 0)\end{aligned}$$

$$\psi_1 = 0$$

$$\begin{aligned}\therefore f(z) &= \int (\psi_1 + i \psi_2) dz = \int [0 + i(1 - z)e^{-z}] dz = i \int (1 - z) e^{-z} dz \\ &= i \left[ (1 - z) \frac{e^{-z}}{-1} - (-1) \frac{e^{-z}}{(-1)^2} + 0 \right] + c_1 \\ &= i \left[ -(1 - z)e^{-z} + e^{-z} \right] + c_1 = i \left[ -\cancel{e^{-z}} + z e^{-z} + \cancel{e^{-z}} \right] + c_1 = i z e^{-z} + c_1\end{aligned}$$

$f(z) = i z e^{-z} + c_1$  is the required analytic function.

### Problem 2

Construct the analytic function by the Milne-Thomson method where its real part is  $u = \frac{2 \sin 2x}{e^{2y} + e^{-2y} - 2 \cos 2x}$ .

**Solution.** Given

$$u = \frac{2 \sin 2x}{e^{2y} + e^{-2y} - 2 \cos 2x} = \frac{2 \sin 2x}{2 \cosh 2y - 2 \cos 2x} \implies u = \frac{\sin 2x}{\cosh 2y - \cos 2x}$$

Now

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{(\cosh 2y - \cos 2x)(2 \cos 2x) - \sin 2x (2 \sin 2x)}{(\cosh 2y - \cos 2x)^2} \\ &= \frac{2 \cosh 2y \cos 2x - 2 \cos^2 2x - 2 \sin^2 2x}{(\cosh 2y - \cos 2x)^2} \\ \frac{\partial u}{\partial x} &= \frac{2 \cosh 2y \cos 2x - 2}{(\cosh 2y - \cos 2x)^2} \\ \phi_1 &= \frac{\partial u}{\partial x}(z, 0) = \frac{2 \cos 2z - 2}{(1 - \cos 2z)^2} \quad (\because \cosh 0 = 1) \\ &= \frac{-2(1 - \cos 2z)}{(1 - \cos 2z)^2} = \frac{-2}{1 - \cos 2z} \\ &= \frac{-2}{2 \sin^2 z} = -\frac{1}{\sin^2 z}\end{aligned}$$

$$\phi_1 = -\operatorname{cosec}^2 z$$

$$\begin{aligned}\frac{\partial u}{\partial y} &= \frac{(\cosh 2y - \cos 2x)(0) - \sin 2x (2 \sinh 2y - 0)}{(\cosh 2y - \cos 2x)^2} = \frac{-2 \sinh 2y \sin 2x}{(\cosh 2y - \cos 2x)^2} \\ \phi_2 &= \frac{\partial u}{\partial y}(z, 0) = 0 \quad (\because \sinh 0 = 0)\end{aligned}$$

$$f(z) = \int (\phi_1 - i \phi_2) dz = \int (-\operatorname{cosec}^2 z - i(0)) dz = -\int \operatorname{cosec}^2 z dz$$

$$f(z) = \cot z + c$$

### Problem 3

Construct the analytic function by the Milne–Thomson method where  $u = \frac{\sin 2x}{\cos 2x + \cosh 2y}$ .

**Solution.** Given  $u = \frac{\sin 2x}{\cos 2x + \cosh 2y}$ .

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{2 \cos 2x [\cos 2x + \cosh 2y] - \sin 2x (-2 \sin 2x)}{(\cos 2x + \cosh 2y)^2} = \frac{2 \cos 2x [\cos 2x + \cosh 2y] + 2 \sin^2 2x}{(\cos 2x + \cosh 2y)^2} \\ &= \frac{2 \cos^2 2x + 2 \cos 2x \cosh 2y + 2 \sin^2 2x}{(\cos 2x + \cosh 2y)^2} = \frac{2 [\cos^2 2x + \sin^2 2x] + 2 \cos 2x \cosh 2y}{(\cos 2x + \cosh 2y)^2} \\ \frac{\partial u}{\partial x} &= \frac{2 + 2 \cos 2x \cosh 2y}{(\cos 2x + \cosh 2y)^2} \\ \frac{\partial u}{\partial x}(z, 0) &= \frac{2 + 2 \cos 2z \cosh 0}{[\cos 2z + \cosh 0]^2} = \frac{2 + 2 \cos 2z}{[\cos 2z + 1]^2} = \frac{2 [1 + \cancel{\cos 2z}]}{[1 + \cos 2z]^2} = \frac{2}{1 + \cos 2z} \\ &= \frac{2}{2 \cos^2 z} = \frac{1}{\cos^2 z} = \sec^2 z\end{aligned}$$

$$\phi_1 = \frac{\partial u}{\partial x}(z, 0) = \sec^2 z$$

$$\begin{aligned}\frac{\partial u}{\partial y} &= \frac{(0)(\cos 2x + \cosh 2y) - (\sin 2x)(2 \sinh 2y)}{(\cos 2x + \cosh 2y)^2} = \frac{-2 \sin 2x \sinh 2y}{(\cos 2x + \cosh 2y)^2} \\ \frac{\partial u}{\partial y}(z, 0) &= 0 \quad (\because \sinh 0 = 0)\end{aligned}$$

$$\phi_2 = \frac{\partial u}{\partial y}(z, 0) = 0$$

$$f(z) = \int (\phi_1 - i \phi_2) dz = \int (\sec^2 z - i(0)) dz$$

$$f(z) = \tan z + c$$

### Note on the question

The handwritten notes write the denominator as  $\cos 2x + \cos 2y$ . With  $\cos 2y$  the given  $u$  is not harmonic, so it cannot be the real part of any analytic function. The standard form of this question has  $\cosh 2y$ , and then  $\operatorname{Re}(\tan z) = \frac{\sin 2x}{\cos 2x + \cosh 2y}$  exactly. (Both versions give the same  $\phi_1, \phi_2$  at  $(z, 0)$ , which is why the method still produced  $\tan z$ .)

### Problem 4

Construct the analytic function  $f(z) = u + iv$  by the Milne–Thomson method, given  $v = \frac{x - y}{x^2 + y^2}$ .

**Solution.** Given  $v = \frac{x - y}{x^2 + y^2}$ .

$$\frac{\partial v}{\partial x} = \frac{(1)(x^2 + y^2) - (x - y)(2x)}{(x^2 + y^2)^2} = \frac{x^2 + y^2 - 2x^2 + 2xy}{(x^2 + y^2)^2} = \frac{-x^2 + y^2 + 2xy}{(x^2 + y^2)^2}$$

$$\frac{\partial v}{\partial x}(z, 0) = \frac{-z^2}{(z^2)^2}$$

$$\psi_2 = -\frac{1}{z^2}$$

$$\frac{\partial v}{\partial y} = \frac{(-1)(x^2 + y^2) - (x - y)(2y)}{(x^2 + y^2)^2} = \frac{-x^2 - y^2 - 2xy + 2y^2}{(x^2 + y^2)^2} = \frac{-x^2 + y^2 - 2xy}{(x^2 + y^2)^2}$$

$$\frac{\partial v}{\partial y}(z, 0) = \frac{-z^2}{(z^2)^2}$$

$$\psi_1 = -\frac{1}{z^2}$$

$$f(z) = \int (\psi_1 + i \psi_2) dz = \int \left[ \left( -\frac{1}{z^2} \right) + i \left( -\frac{1}{z^2} \right) \right] dz = -(1 + i) \int z^{-2} dz$$

$$= -(1 + i) \left( -\frac{1}{z} \right) + c$$

$$f(z) = \frac{1 + i}{z} + c$$

### Problems 5 & 6

Construct the analytic function  $f(z) = u + iv$  by the Milne-Thomson method, given

(5)  $v = e^x(x \sin y + y \cos y)$

(6)  $u = \frac{1}{2} \log_e(x^2 + y^2)$

**5) Solution.** Given  $v = e^x(x \sin y + y \cos y)$ .

$$\frac{\partial v}{\partial y} = e^x [x \cos y + (1) \cos y - y \sin y] = e^x [x \cos y + \cos y - y \sin y]$$

$$\psi_1 = \frac{\partial v}{\partial y}(z, 0) = e^z(z(1) + (1) - 0) = e^z(1 + z)$$

$$\frac{\partial v}{\partial x} = e^x(\sin y) + e^x(x \sin y + y \cos y)$$

$$\psi_2 = \frac{\partial v}{\partial x}(z, 0) = e^z(0) + e^z(z(0) + 0) = 0$$

$$f(z) = \int (\psi_1 + i \psi_2) dz = \int [e^z + z e^z] dz = e^z + [z e^z - e^z] + c$$

$$f(z) = z e^z + c$$

**6) Solution.** Given  $u = \frac{1}{2} \log_e(x^2 + y^2)$ .

$$\frac{\partial u}{\partial x} = \frac{1}{2} \left( \frac{1}{x^2 + y^2} \right) (2x),$$

$$\phi_1 = \frac{\partial u}{\partial x}(z, 0) = \frac{1}{2} \left[ \frac{1}{z^2 + 0} \times 2z \right] = \frac{1}{z}$$

$$\frac{\partial u}{\partial y} = \frac{1}{2} \left( \frac{1}{x^2 + y^2} \right) (2y),$$

$$\phi_2 = \frac{\partial u}{\partial y}(z, 0) = \frac{1}{2z^2} (2(0)) = 0$$

$$f(z) = \int (\phi_1 - i \phi_2) dz = \int \frac{1}{z} dz$$

$$f(z) = \log z + c$$

### 6.1 When a linear combination of $u$ and $v$ is given

Class of 09/09/26

#### Problem 1

Construct the analytic function  $f(z)$  by the Milne–Thomson method where

$$2u + v = e^{2x} [(2x + y) \cos 2y + (x - 2y) \sin 2y].$$

**Solution.** Given  $2u + v = e^{2x} [(2x + y) \cos 2y + (x - 2y) \sin 2y]$ .

Let  $f(z) = u + iv$  be the required analytic function. — (1)

$$\begin{aligned} (1) \times 2 &\implies 2f(z) = 2u + i2v \\ (1) \times (-i) &\implies -if(z) = -iu + v \end{aligned}$$

Adding the above two equations,

$$(2 - i)f(z) = (2u + v) + i(2v - u) \implies F(z) = U + iV \quad \text{— (2)}$$

where  $F(z) = (2 - i)f(z)$ ,  $U = 2u + v$  and  $V = 2v - u$ .

So in the above analytic function  $F(z) = U + iV$ , we have its real part

$$U = 2u + v = e^{2x} [(2x + y) \cos 2y + (x - 2y) \sin 2y]$$

$$\begin{aligned} \frac{\partial U}{\partial x} &= 2e^{2x} [(2x + y) \cos 2y + (x - 2y) \sin 2y] + e^{2x} [2 \cos 2y + \sin 2y] \\ \phi_1 = \frac{\partial U}{\partial x}(z, 0) &= 2e^{2z} [2z] + e^{2z} [2] \end{aligned}$$

$$\phi_1 = 2e^{2z}(2z + 1)$$

$$\begin{aligned} \frac{\partial U}{\partial y} &= e^{2x} [(2x + y)(-\sin 2y)(2) + \cos 2y + (x - 2y)(2 \cos 2y) + (-2) \sin 2y] \\ \phi_2 = \frac{\partial U}{\partial y}(z, 0) &= e^{2z} [1 + z(2)] \end{aligned}$$

$$\phi_2 = (2z + 1)e^{2z}$$

$$\begin{aligned} \therefore F(z) &= \int (\phi_1 - i\phi_2) dz = \int [2e^{2z}(2z + 1) - i(2z + 1)e^{2z}] dz + c \\ &= \int (2 - i)(2z + 1) e^{2z} dz + c = (2 - i) \int (2z + 1) e^{2z} dz + c \\ &= (2 - i) \left[ (2z + 1) \left( \frac{e^{2z}}{2} \right) - (2) \left( \frac{e^{2z}}{4} \right) + 0 \right] + c \\ &= \frac{2 - i}{2} [e^{2z}(2z + 1 - 1)] + c = \frac{2 - i}{2} [2ze^{2z}] + c \\ F(z) &= (2 - i)ze^{2z} + c \end{aligned}$$

But  $F(z) = (2 - i)f(z)$ :

$$(2 - i)f(z) = (2 - i)ze^{2z} + c \implies f(z) = ze^{2z} + \frac{c}{2 - i}$$

$$f(z) = z e^{2z} + k, \quad \text{where } k \text{ is a complex constant.}$$

**Problem 2**

Construct the analytic function  $f(z) = u + iv$ , given  $3u + 2v = y^2 - x^2 + 16xy$ .

**Solution.** Given  $3u + 2v = y^2 - x^2 + 16xy$ .

Let  $f(z) = u + iv$  be an analytic function. — (1)

$$(1) \times 3 \implies 3f(z) = 3u + i3v$$

$$(1) \times (-2i) \implies -2if(z) = -2iu + 2v$$

Add  $\implies (3 - 2i)f(z) = (3u + 2v) + i(3v - 2u)$ , i.e.  $F(z) = U + iV$ , where  $F(z) = (3 - 2i)f(z)$  and  $U = 3u + 2v$  is given:

$$U = y^2 - x^2 + 16xy$$

$$\begin{array}{c|c} \frac{\partial U}{\partial x} = -2x + 16y & \frac{\partial U}{\partial y} = 2y + 16x \\ \phi_1 = \frac{\partial U}{\partial x}(z, 0) = -2z & \phi_2 = \frac{\partial U}{\partial y}(z, 0) = 16z \end{array}$$

$$F(z) = \int (\phi_1 - i\phi_2) dz = \int \phi_1 dz - i \int \phi_2 dz + c = \int -2z dz - i \int 16z dz + c$$

$$= (-2 - 16i) \int z dz + c = (-2 - 16i) \left( \frac{z^2}{2} \right) + c$$

$$F(z) = -(1 + 8i)z^2 + c$$

$$(3 - 2i)f(z) = -(1 + 8i)z^2 + c$$

$$\begin{aligned} f(z) &= -\frac{1 + 8i}{3 - 2i} z^2 + \frac{c}{3 - 2i} = -z^2 \left[ \frac{1 + 8i}{3 - 2i} \times \frac{3 + 2i}{3 + 2i} \right] + k \\ &= -z^2 \left[ \frac{3 + 2i + 24i - 16}{9 + 4} \right] + k = -z^2 \left[ \frac{26i - 13}{13} \right] + k \end{aligned}$$

$$f(z) = (1 - 2i)z^2 + k$$

**Problem 3**

Construct the analytic function  $f(z) = u + iv$  where

$$(i) \quad u + v = \frac{x}{x^2 + y^2}$$

$$(ii) \quad u - v = \frac{\cos x + \sin x - e^{-y}}{2 \cos x - (e^y + e^{-y})}, \quad \text{given } f\left(\frac{\pi}{2}\right) = 0.$$

**(i) Solution.** Given  $u + v = \frac{x}{x^2 + y^2}$ .

Let  $f(z) = u + iv$  be an analytic function. — (1)

$$(1) \times (-i) \implies -if(z) = -iu + v \quad \text{— (2)}$$

Add (1) and (2):

$$(1 - i)f(z) = (u + v) + i(v - u) \implies F(z) = U + iV$$

where  $F(z) = (1-i)f(z)$  and  $U = u + v = \frac{x}{x^2 + y^2}$ .

$$\frac{\partial U}{\partial x} = \frac{(x^2 + y^2)(1) - x(2x)}{(x^2 + y^2)^2} = \frac{(x^2 + y^2) - 2x^2}{(x^2 + y^2)^2}$$

$$\frac{\partial U}{\partial x}(z, 0) = \frac{z^2 - 2z^2}{(z^2)^2} = \frac{-z^2}{z^4}$$

$$\phi_1 = \frac{\partial U}{\partial x}(z, 0) = -\frac{1}{z^2}$$

$$\frac{\partial U}{\partial y} = \frac{(x^2 + y^2)(0) - x(2y)}{(x^2 + y^2)^2} = \frac{-2xy}{(x^2 + y^2)^2}, \quad \frac{\partial U}{\partial y}(z, 0) = 0$$

$$\phi_2 = \frac{\partial U}{\partial y}(z, 0) = 0$$

$$F(z) = \int (\phi_1 - i\phi_2) dz = \int \left(-\frac{1}{z^2}\right) dz + c \implies F(z) = \frac{1}{z} + c$$

$$(1-i)f(z) = \frac{1}{z} + c \implies f(z) = \frac{1}{z(1-i)} + \frac{c}{1-i} = \frac{1+i}{z(1-i)(1+i)} + k$$

$$f(z) = \frac{1+i}{2z} + k$$

**(ii) Solution.** Given  $u - v = \frac{\cos x + \sin x - e^{-y}}{2 \cos x - (e^y + e^{-y})}$  and  $f(\pi/2) = 0$ .

Let  $f(z) = u + iv$  be an analytic function. — (1)

$$(1) \times i \implies if(z) = iu - v \quad \text{— (2)}$$

Add:

$$f(z) + if(z) = (u - v) + i(v + u) \implies (1+i)f(z) = (u - v) + i(u + v) \implies F(z) = U + iV,$$

where  $F(z) = (1+i)f(z)$  and

$$U = u - v = \frac{\cos x + \sin x - e^{-y}}{2 \cos x - (e^y + e^{-y})} = \frac{\cos x + \sin x - e^{-y}}{2(\cos x - \cosh y)} \quad (\because e^y + e^{-y} = 2 \cosh y)$$

$$\frac{\partial U}{\partial x} = \frac{2(\cos x - \cosh y)(-\sin x + \cos x) - (\cos x + \sin x - e^{-y})(-2 \sin x)}{(2(\cos x - \cosh y))^2}$$

$$\phi_1 = \frac{\partial U}{\partial x}(z, 0) = \frac{(2 \cos z - 2)(-\sin z + \cos z) - (\cos z + \sin z - 1)(-2 \sin z)}{4(\cos z - 1)^2}$$

$$= \frac{2[\cos^2 z - \cos z - \sin z \cos z + \sin z + \sin z \cos z + \sin^2 z - \sin z]}{4(\cos z - 1)^2}$$

$$= \frac{2(1 - \cos z)}{4(1 - \cos z)^2} = \frac{1}{2(1 - \cos z)} = \frac{1}{4 \sin^2(z/2)}$$

$$\phi_1 = \frac{1}{4} \operatorname{cosec}^2\left(\frac{z}{2}\right)$$

$$\frac{\partial U}{\partial y} = \frac{2(\cos x - \cosh y)(e^{-y}) - (\cos x + \sin x - e^{-y})(-2 \sinh y)}{4(\cos x - \cosh y)^2}$$

$$\phi_2 = \frac{\partial U}{\partial y}(z, 0) = \frac{2(\cos z - 1)(1) - 0}{4(\cos z - 1)^2} = \frac{1}{2(\cos z - 1)} = -\frac{1}{2(1 - \cos z)} = -\frac{1}{4\sin^2(z/2)}$$

$$\phi_2 = -\frac{1}{4} \operatorname{cosec}^2\left(\frac{z}{2}\right)$$

$$\begin{aligned} F(z) &= \int (\phi_1 - i\phi_2) dz = \int \left(\frac{1}{4} \operatorname{cosec}^2\left(\frac{z}{2}\right) + i\frac{1}{4} \operatorname{cosec}^2\left(\frac{z}{2}\right)\right) dz = \left(\frac{1+i}{4}\right) \int \operatorname{cosec}^2\left(\frac{z}{2}\right) dz \\ &= \left(\frac{1+i}{4}\right) \left[\frac{-\cot(z/2)}{1/2}\right] + c \end{aligned}$$

$$F(z) = -\left(\frac{1+i}{2}\right) \cot\left(\frac{z}{2}\right) + c$$

$$(1+i)f(z) = -\left(\frac{1+i}{2}\right) \cot\left(\frac{z}{2}\right) + c \implies f(z) = -\frac{1}{2} \cot\left(\frac{z}{2}\right) + k$$

Given  $f(\pi/2) = 0$ :

$$-\frac{1}{2} \cot\left(\frac{\pi}{4}\right) + k = 0 \implies -\frac{1}{2}(1) + k = 0 \implies k = \frac{1}{2}$$

$$f(z) = \frac{1}{2} \left[1 - \cot\left(\frac{z}{2}\right)\right]$$

### Check

In the handwritten notes the signs of  $\phi_1$  and  $\phi_2$  were both interchanged, which led to  $f(z) = \frac{1}{2} [\cot(z/2) - 1]$ . With the signs as worked above,  $\operatorname{Re} f - \operatorname{Im} f$  reproduces the given  $u - v$  exactly, so the answer is  $f(z) = \frac{1}{2} [1 - \cot(z/2)]$ .

## 7 Problems on Analytic and Harmonic Functions

Class of 10/09/26

### Problem 1

If  $f(z) = u + iv$  is an analytic function, prove that (a)  $v - iu$  is also an analytic function; (b)  $v + iu$  is not an analytic function.

**Solution.**  $f(z) = u + iv$  is an analytic function, so its real part  $u$  and imaginary part  $v$  must satisfy the Cauchy–Riemann (C–R) equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \implies u_x = v_y \quad \text{--- (1)} \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \implies u_y = -v_x \quad \text{--- (2)}$$

(a) Let  $F(z) = v - iu = -i(u + iv)$ . Since  $f(z) = u + iv$ ,

$$F(z) = -if(z)$$

which is a constant multiple of an analytic function.

$\therefore v - iu$  is an analytic function.

(b) Let  $F(z) = v + iu = U + iV$ , with  $U = v$  and  $V = u$ . If  $F$  were analytic, the C–R equations  $U_x = V_y$ ,  $U_y = -V_x$  would give

$$v_x = u_y \quad \text{--- (3)} \qquad \text{and} \qquad v_y = -u_x \quad \text{--- (4)}$$

From (2):  $u_y = -v_x$ , and by (3)

$$\begin{aligned} v_x &= -v_x \\ 2v_x &= 0 \implies v_x = 0 \end{aligned}$$

From (1):  $u_x = v_y$ , and by (4)

$$\begin{aligned} v_y &= -v_y \\ 2v_y &= 0 \implies v_y = 0 \end{aligned}$$

This would force  $u$  and  $v$  to be constants. But  $u$  and  $v$  are not constants, so the C–R equations are not satisfied.

$\therefore v + iu$  is not an analytic function.

### Problem 2

Find the points at which the C–R equations are satisfied by the function  $xy^2 + i yx^2$ .

**Solution.** Let  $f(z) = u + iv$ . Given  $f(z) = xy^2 + i yx^2$ , so  $u = xy^2$ ,  $v = x^2y$ .

$$\begin{aligned} \frac{\partial u}{\partial x} &= y^2 \\ \frac{\partial u}{\partial y} &= 2xy \end{aligned}$$

$$\begin{aligned} \frac{\partial v}{\partial x} &= 2xy \\ \frac{\partial v}{\partial y} &= x^2 \end{aligned}$$

By the C–R equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \implies y^2 = x^2 \quad \text{--- (1)}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \implies 2xy = -2xy \implies 4xy = 0 \implies xy = 0 \quad \text{--- (2)}$$

From (2),  $x = 0$  or  $y = 0$ . If  $x = 0$ , (1) gives  $y = 0$ ; if  $y = 0$ , (1) gives  $x = 0$ .

The C–R equations are satisfied only at the point  $(0, 0)$ , i.e. at  $z = 0$ .  
 $\therefore$  The function is not analytic anywhere.

### Problem 3

Determine the value of  $p$  such that  $f(z) = \frac{1}{2} \log(x^2 + y^2) + i \tan^{-1}\left(\frac{px}{y}\right)$  is an analytic function.

**Solution.** Given  $f(z)$  is analytic, with  $u = \frac{1}{2} \log(x^2 + y^2)$  and  $v = \tan^{-1}\left(\frac{px}{y}\right)$ .

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{1}{2} \cdot \frac{2x}{x^2 + y^2} = \frac{x}{x^2 + y^2} \\ \frac{\partial u}{\partial y} &= \frac{1}{2} \cdot \frac{2y}{x^2 + y^2} = \frac{y}{x^2 + y^2} \end{aligned}$$

$$\begin{aligned} \frac{\partial v}{\partial x} &= \frac{1}{1 + \frac{p^2 x^2}{y^2}} \cdot \frac{p}{y} = \frac{py}{y^2 + p^2 x^2} \\ \frac{\partial v}{\partial y} &= \frac{1}{1 + \frac{p^2 x^2}{y^2}} \left( \frac{-px}{y^2} \right) = \frac{-px}{y^2 + p^2 x^2} \end{aligned}$$

By the C–R equations,

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y} \implies \frac{x}{x^2 + y^2} = \frac{-px}{y^2 + p^2 x^2} \\ &\implies y^2 + p^2 x^2 = -p(x^2 + y^2) = -px^2 - py^2 \\ &\implies p^2 x^2 + px^2 + y^2 + py^2 = 0 \\ &\implies x^2(p^2 + p) + y^2(1 + p) = 0 \end{aligned}$$

$$p^2 + p = 0 \Rightarrow p(p+1) = 0 \Rightarrow p = 0 \text{ or } p = -1$$

$$1 + p = 0 \Rightarrow p = -1$$

Check the second equation with  $p = -1$ :

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \Rightarrow \frac{y}{x^2 + y^2} = -\frac{(-1)y}{y^2 + (-1)^2 x^2} = \frac{y}{y^2 + x^2} \quad \checkmark$$

C-R equations are satisfied;  $p = -1$  makes  $f(z)$  an analytic function.

#### Problem 4

If  $f(z) = u + iv = \cosh z$  is an analytic function, show that the curves  $u = \text{const}$  and  $v = \text{const}$  are orthogonal.

**Solution.** Given  $f(z) = u + iv = \cosh z$ .

$$\cosh z = \cosh(x + iy) = \cosh x \cos y + i \sinh x \sin y$$

$$u = \cosh x \cos y$$

$$\frac{\partial u}{\partial x} = \sinh x \cos y$$

$$\frac{\partial u}{\partial y} = -\cosh x \sin y$$

$$v = \sinh x \sin y$$

$$\frac{\partial v}{\partial x} = \cosh x \sin y$$

$$\frac{\partial v}{\partial y} = \sinh x \cos y$$

By the C-R equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} : \sinh x \cos y = \sinh x \cos y; \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} : -\cosh x \sin y = -\cosh x \sin y.$$

Consider the family of curves  $u(x, y) = c_1$ :

$$0 = du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \Rightarrow \frac{\partial u}{\partial y} dy = -\frac{\partial u}{\partial x} dx \Rightarrow \frac{dy}{dx} = \frac{-\partial u / \partial x}{\partial u / \partial y} \Rightarrow m_1 = -\frac{u_x}{u_y}$$

and the family  $v(x, y) = c_2$ :

$$0 = dv = \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy \Rightarrow \frac{\partial v}{\partial y} dy = -\frac{\partial v}{\partial x} dx \Rightarrow \frac{dy}{dx} = -\frac{v_x}{v_y} \Rightarrow m_2 = -\frac{v_x}{v_y}$$

$$m_1 \cdot m_2 = \left(-\frac{u_x}{u_y}\right) \left(-\frac{v_x}{v_y}\right) = \frac{u_x v_x}{u_y v_y} = \frac{v_y \cdot v_x}{(-v_x) \cdot v_y} = -1 \quad (\because u_x = v_y, u_y = -v_x)$$

$m_1 \cdot m_2 = -1$ : the two families of curves  $u = \text{const}$  and  $v = \text{const}$  cut each other at right angles.

#### Problem 5

Find  $a, b$  if  $f(z) = (x^2 - 2xy + ay^2) + i(bx^2 - y^2 + 2xy)$  is an analytic function, given  $a$  and  $b$  are real, and also find  $f(z)$  in terms of  $z$ .

**Solution.** Given  $f(z) = u + iv = (x^2 - 2xy + ay^2) + i(bx^2 - y^2 + 2xy)$ .

$$\begin{aligned}
 u &= x^2 - 2xy + ay^2 \\
 \frac{\partial u}{\partial x} &= 2x - 2y \\
 \frac{\partial u}{\partial y} &= -2x + 2ay
 \end{aligned}$$

$$\begin{aligned}
 v &= bx^2 - y^2 + 2xy \\
 \frac{\partial v}{\partial x} &= 2bx + 2y \\
 \frac{\partial v}{\partial y} &= -2y + 2x
 \end{aligned}$$

By the C-R equations,

$$\begin{aligned}
 \frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y} \implies 2x - 2y = -2y + 2x \\
 \frac{\partial u}{\partial y} &= -\frac{\partial v}{\partial x} \implies -2x + 2ay = -(2bx + 2y) = -2bx - 2y
 \end{aligned}$$

Comparing the  $x$  and  $y$  terms:

$$-2 = -2b \implies \boxed{b = 1} \qquad 2a = -2 \implies \boxed{a = -1}$$

Substituting  $a$  in  $u$ :  $u = x^2 - 2xy - y^2$ .

$$\begin{aligned}
 \frac{\partial u}{\partial x} &= 2x - 2y, & \phi_1 &= \frac{\partial u}{\partial x}(z, 0) = 2(z) - 2(0) = 2z \\
 \frac{\partial u}{\partial y} &= -2x - 2y, & \phi_2 &= \frac{\partial u}{\partial y}(z, 0) = -2(z) - 2(0) = -2z
 \end{aligned}$$

$$\begin{aligned}
 f(z) &= \int (\phi_1 - i\phi_2) dz = \int (2z - i(-2z)) dz = \int (2z + 2iz) dz \\
 &= (2 + 2i) \int z dz = (2 + 2i) \left( \frac{z^2}{2} \right) + c
 \end{aligned}$$

$$f(z) = (1 + i)z^2 + c$$

### Problem 6

If  $u$  is a harmonic function, show that  $u^2$  is not a harmonic function unless  $u$  is a constant.

**Solution.** Given  $u$  is a harmonic function:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{--- (1)}$$

To check whether  $u^2$  is harmonic, compute  $\left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) (u^2)$ .

Differentiating w.r.t.  $x$ :

$$\begin{aligned}
 \frac{\partial}{\partial x}(u^2) &= 2u \frac{\partial u}{\partial x} \\
 \frac{\partial^2}{\partial x^2}(u^2) &= 2u \frac{\partial^2 u}{\partial x^2} + 2 \left( \frac{\partial u}{\partial x} \right)^2 \quad \text{--- (2)}
 \end{aligned}$$

Differentiating w.r.t.  $y$ :

$$\begin{aligned}
 \frac{\partial}{\partial y}(u^2) &= 2u \frac{\partial u}{\partial y} \\
 \frac{\partial^2}{\partial y^2}(u^2) &= 2u \frac{\partial^2 u}{\partial y^2} + 2 \left( \frac{\partial u}{\partial y} \right)^2 \quad \text{--- (3)}
 \end{aligned}$$

(2) + (3)  $\implies$

$$\frac{\partial^2}{\partial x^2}(u^2) + \frac{\partial^2}{\partial y^2}(u^2) = 2u \left( \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + 2 \left[ \left( \frac{\partial u}{\partial x} \right)^2 + \left( \frac{\partial u}{\partial y} \right)^2 \right]$$

Substituting (1):

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)u^2 = 0 + 2\left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2\right]$$

For  $u^2$  to be harmonic we need  $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)(u^2) = 0$ :

$$\Rightarrow 2\left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2\right] = 0 \Rightarrow \frac{\partial u}{\partial x} = 0 \text{ and } \frac{\partial u}{\partial y} = 0,$$

so  $u$  must be a constant function.

$u^2$  is not harmonic unless  $u$  is a constant.

### Problem 7

Find whether  $u = y + e^x \cos y$  can be the real part of an analytic function.

**Solution.** Given  $u = y + e^x \cos y$ .

$$\begin{aligned}\frac{\partial u}{\partial x} &= e^x \cos y \\ \frac{\partial^2 u}{\partial x^2} &= e^x \cos y\end{aligned}$$

$$\begin{aligned}\frac{\partial u}{\partial y} &= 1 + e^x(-\sin y) = 1 - e^x \sin y \\ \frac{\partial^2 u}{\partial y^2} &= -e^x \cos y\end{aligned}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = e^x \cos y + (-e^x \cos y) = 0$$

$u$  satisfies Laplace's equation, so  $u$  is harmonic.

$\therefore u = y + e^x \cos y$  can be the real part of an analytic function.

### Problem 8

If  $u = x^2 - y^2$  and  $v = \frac{-2y}{x^2 + y^2}$ , prove that  $u$  and  $v$  are harmonic functions, but they are not harmonic conjugates of each other.

**Solution.** Given  $u = x^2 - y^2$  and  $v = \frac{-2y}{x^2 + y^2}$ .

$$\begin{array}{l|l}
\frac{\partial u}{\partial x} = 2x & \frac{\partial v}{\partial x} = -2y \cdot \frac{-1}{(x^2 + y^2)^2} (2x) = \frac{4xy}{(x^2 + y^2)^2} \\
\frac{\partial^2 u}{\partial x^2} = 2 & \frac{\partial^2 v}{\partial x^2} = \frac{(x^2 + y^2)^2 (4y) - 4xy \cdot 2(x^2 + y^2)(2x)}{(x^2 + y^2)^4} \\
\frac{\partial u}{\partial y} = -2y & = \frac{4y^3 - 12x^2y}{(x^2 + y^2)^3} \\
\frac{\partial^2 u}{\partial y^2} = -2 & \frac{\partial v}{\partial y} = \frac{(x^2 + y^2)(-2) - (-2y)(2y)}{(x^2 + y^2)^2} = \frac{-2x^2 + 2y^2}{(x^2 + y^2)^2} \\
\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2 + (-2) = 0 & \frac{\partial^2 v}{\partial y^2} = \frac{(x^2 + y^2)^2 (4y) - (-2x^2 + 2y^2) \cdot 2(x^2 + y^2)(2y)}{(x^2 + y^2)^4} \\
\therefore u \text{ is a harmonic function.} & = \frac{12x^2y - 4y^3}{(x^2 + y^2)^3} \\
& \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = \frac{4y^3 - 12x^2y + 12x^2y - 4y^3}{(x^2 + y^2)^3} = 0
\end{array}$$

$\therefore v$  is also a harmonic function.

To check whether  $u$  and  $v$  are harmonic conjugates, they must satisfy the C-R equations  $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$  and  $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ . But

$$\frac{\partial u}{\partial x} = 2x \neq \frac{2y^2 - 2x^2}{(x^2 + y^2)^2} = \frac{\partial v}{\partial y} \quad \therefore \frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$$

C-R equations are not satisfied;  $u$  and  $v$  are not harmonic conjugates.

### Problem 9

If  $f(z) = u(r, \theta) + i v(r, \theta)$  is an analytic function, prove that

$$r^2 \frac{\partial^2 u}{\partial r^2} + r \frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial \theta^2} = 0.$$

(This equation is called **Laplace's equation in polar form**.)

### Proof

Given  $f(z) = u + iv$  is analytic, so the polar C-R equations hold:

$$\begin{array}{l}
\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} \\
\frac{\partial u}{\partial \theta} = -r \frac{\partial v}{\partial r}
\end{array}$$

$$\begin{array}{l}
\frac{\partial v}{\partial \theta} = r \frac{\partial u}{\partial r} \\
\frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}
\end{array}$$

$$\begin{aligned}
\frac{\partial^2 v}{\partial r \partial \theta} &= \frac{\partial}{\partial r} \left( r \frac{\partial u}{\partial r} \right) = (1) \frac{\partial u}{\partial r} + r \frac{\partial^2 u}{\partial r^2} \\
\frac{\partial^2 v}{\partial \theta \partial r} &= \frac{\partial}{\partial \theta} \left( -\frac{1}{r} \frac{\partial u}{\partial \theta} \right) = -\frac{1}{r} \frac{\partial^2 u}{\partial \theta^2}
\end{aligned}$$

Since  $\frac{\partial^2 v}{\partial r \partial \theta} = \frac{\partial^2 v}{\partial \theta \partial r}$ ,

$$\frac{\partial u}{\partial r} + r \frac{\partial^2 u}{\partial r^2} = -\frac{1}{r} \frac{\partial^2 u}{\partial \theta^2}$$

$$r \frac{\partial u}{\partial r} + r^2 \frac{\partial^2 u}{\partial r^2} = -\frac{\partial^2 u}{\partial \theta^2}$$

$$r^2 \frac{\partial^2 u}{\partial r^2} + r \frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial \theta^2} = 0$$

Hence proved.